**7034SSL- Business Statistics**

**CW1 Group Report**

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# Contents

1. Introduction..... **Error! Bookmark not defined.**
2. Literature Review Using 5 peer-reviewed case study: .... **Error! Bookmark not defined.**
   2.1 Environmental, Social, and Governance aspects in the Ranking of Companies**:**...**Error! Bookmark not defined.**
   2.2 Technology Acceptance and Business Outcomes: A Longitudinal Analysis:...……..3
   2.3 The Perception of Stakeholders and Its Effect on Corporate Rankings……...4
   2.4 Comparative Analysis of Industry-Specific Challenges in Corporate Rankings:……4
   2.5 Cross-Sector Comparisons: A Tool for Assessing Company Rankings……5
3. Variable Analysis .....5
4. Steps used for data cleaning.....6
5. Descriptive Statistic measurements in Forbes Companies Ranking……..7
   5.1 Descriptive statistics for adjusted revenue.....7
   5.2 Descriptive statistics for adjusted profit .....8
   5.3 Descriptive statistics for adjusted asset.....10
   5.4 Descriptive statistics for adjusted market value.....11
   5.5 Descriptive statistics for adjusted total employees .....13
6. Result .....14
7. Discussion……15

References.....17

Self Reflection .....20

**List of tables:**

Table 1: Descriptive Statistics for Adjusted Revenue ..... 7
Table 2: Descriptive Statistics for Adjusted Profit ..... 8
Table 3: Descriptive Statistics for Adjusted Assets ..... 10
Table 4: Descriptive Statistics for Adjusted Market Value ..... 11
Table 5: Descriptive Statistics for Total Employees ..... 13

**List of Figures:**

Figure 1: A Graphical representation of Descriptive Statistics for Adjusted Revenue. ..... 7
Figure 2: A Graphical representation of Descriptive Statistics for Adjusted Profit ..... 9
Figure 3: A Graphical representation of Descriptive Statistics for Adjusted Assets. ..... 10
Figure 4: A Graphical representation of Descriptive Statistics for Adjusted Market Value .... 12
Figure 5: A Graphical representation of Descriptive Statistics for Total Employees. ..... 13

## 1. Introduction:

Company rankings based on a composite score of revenue, profit, assets, and market value. The energetic focal point of Company Rankings offers a see into the ever-evolving structure of universal trade. The extreme compilation of the greatest freely exchanged firms around the world, gives an up-to-date see of the world's most powerful companies, exhibiting their adaptability and grit within the confront of moving advertise flow. The list's steady alter reflects the versatility of commerce victory, appearing how businesses overcome deterrents, welcome alter, and build up themselves as front-runners in their individual areas. Ranking encapsulates the energetic nature of corporate scenes, from the tech sector's speedy breakthroughs to the recede and stream of money related mammoths. It gives partners, financial specialists, and devotees with an all-encompassing see of the cutting edge commerce environment.

Individual and company rankings offer insight into the dynamic world of personal wealth and entrepreneurship. These rankings highlight the journeys of influential individuals and the fluctuations in their fortunes. Whether featuring young innovators or top employers, It is reveal the resilience and adaptability of businesses and individuals in an ever-evolving global landscape.

The Forbes Global 2000 list ranks the world's largest companies based on four key metrics: sales, profits, assets, and market value. Due to market volatility, the minimum market value required for inclusion in the 2022 list has decreased to $7.6 billion, compared to $8.26 billion in 2021. However, the minimum thresholds for the other three metrics have increased compared to the previous year. This shift reflects the dynamic nature of the global economy and the impact of external factors on corporate performance. Despite the challenges, the companies on the Forbes Global 2000 list continue to demonstrate resilience and adaptability in the face of changing market conditions.

The 2022 Forbes Global 2000 list saw Berkshire Hathaway take the top spot for the first time, ending the Industrial and Commercial Bank of China's nine-year reign at No. 1. The list also revealed a resurgence for big oil companies, as rising oil prices helped ExxonMobil and Shell rebound from low rankings in 2021 to No. 15 and No. 16, respectively. Both companies returned to profitability after experiencing losses the previous year. These shifts demonstrate the ever-changing nature of the global business environment, as companies continuously adapt to new market conditions and aim for growth and prosperity.


- Contreras, I. (2022, May 12). Forbes Global 2000 List 2022: The Top 200. *Forbes*.

## 2. Literature Review About Company Rankings using 5 peer-reviewed studies published in the last five years:

### 2.1. Environmental, Social, and Governance (ESG) Aspects in the Ranking of Companies

According to the study "Environmental, social, and governance (ESG) performance and financial outcomes: Analysing the impact of ESG on financial performance," a company's short-term expenses go up when it incorporates ESG practices. However, their medium- and long-term financial performance can improve. Some researchers believe that the relationship between financially successful companies and corporate social responsibility is not a simple linear one, but rather an inverted U-shaped curve.

Recent studies have found that higher levels of sustainability in exchange-traded funds (ETFs) can lead to better performance during economic downturns, such as the Covid-19 pandemic. Using ANOVA and multivariate regression models, Dugard et al. (2022) analysed the financial returns of ETFs and their Eco-Fund rankings during market crashes. The results showed that in emerging markets, better financial performance was linked to higher environmental, social, and governance (ESG) standards. This suggests that sustainable investments may be more resilient during economic crises.

The COVID-19 pandemic has led to strict regulations and social distancing measures in many countries (You et al., 2022). These measures have caused significant disruptions to businesses, including supply chain interruptions, decreased product demand, and reduced operations due to prevention and control procedures. As a result, businesses have experienced substantial negative economic impacts. The restrictions have had a tangible and severe effect on the global economy, highlighting the challenges posed by the pandemic.

The article "Hotspots and trends of environmental, social and governance (ESG) research: a bibliometric analysis" provides substantial evidence for the relationships between ESG and various aspects of corporate performance. The study shows that companies with poor environmental practices tend to have lower credit ratings and higher yields, indicating a direct connection between environmental performance and financial outcomes. This highlights the importance of incorporating ESG considerations into corporate decision-making and risk management strategies. The research contributes to the growing body of evidence demonstrating the financial materiality of ESG factors.

- Chen, S., Song, Y., & Gao, P. (2023). Environmental, social, and governance (ESG) performance and financial outcomes: Analysing the impact of ESG on financial performance. *Journal of Environmental Management*, *345*, 118829.

- Dugard, P., Todman, J., & Staines, H. (2022). *Approaching multivariate analysis: A practical introduction*. Taylor & Francis.
- You, G., Gan, S., Guo, H., & Dagestani, A. A. (2022). Public opinion spread and guidance strategy under COVID-19: A SIS model analysis. *Axioms*, *11*(6), 296.
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- Zhang, W., Luo, Q., & Liu, S. (2022). Is government regulation a push for corporate environmental performance? Evidence from China. *Economic Analysis and Policy*, *74*, 105-121.


### 2.2. Technology Acceptance and Business Outcomes: A Longitudinal Analysis

The article "Developing a human-organization-technology fit model for information technology adoption in organizations" explores various theories that aim to explain how organizations adopt and integrate new technologies. These theories include the task-technology fit (T-T fit) model, diffusion of innovation (DoI) theory, technology acceptance model (TAM), adaptive structuration theory (AST), technology-organization-environment (TOE) framework, and technology affordance theory (TAT). These theories examine the processes and factors that influence an organization's decision to adopt new technologies, as well as the specific outcomes or impacts of broader technological innovations. The article highlights the importance of understanding the interplay between technology, individuals, and organizations in order to facilitate successful technology adoption and integration.

To further examine the connection between IT capabilities and firm performance, we utilized the construct of IT's strategic role in various industries, as previously mentioned. We believe this can help elucidate the interplay between different industry contexts and IT capabilities. Moreover, the level of IT investment by managers is a critical factor, as they must choose an appropriate portfolio of IT assets and navigate the trade-off between the competitive risk of underinvesting during challenging times and the financial risk of overinvesting. This highlights the importance of strategic IT decision-making in achieving optimal firm performance.


- Xu, J., & Lu, W. (2022). Developing a human-organization-technology fit model for information technology adoption in organizations. *Technology in Society*, *70*, 102010.
- Chae, H., Koh, C. E., & Park, K. O. (2018). Information technology capability and firm performance: Role of industry. *Information & Management, 55(5),* 525–546.

### 2.3. The Perception of Stakeholders and Its Effect on Corporate Rankings

Stakeholder management strategies are gaining prominence due to the substantial impact that stakeholders can have on decision-making processes. One approach to understanding stakeholder influence is the classic demographic approach, which links influence to specific personal characteristics. Scholars have also explored the role of stakeholder influence across various theoretical frameworks and disciplines, with previous research demonstrating its expansion and diversity (Fares et al., 2021). Stakeholder influence can take various forms and hold different meanings for different people, highlighting the complexity of stakeholder management in decision-making contexts.

Accordingly, this article (A multi-level typology for stakeholder influence: A systematic literature review using the structural approach) managers must understand the resource exchange dynamics inherent in stakeholder relationships to accurately model stakeholder influence behaviour. This understanding is crucial for implementing effective stakeholder management practices and making informed decisions. By recognizing the complex interplay of resources and influence among stakeholders, managers can develop more sophisticated and effective strategies for engaging with stakeholders and navigating their impact on decision-making processes. This highlights the importance of a nuanced approach to stakeholder management that accounts for the unique dynamics of different stakeholder relationships.

- Fares, J., Chung, K. S. K., & Abbasi, A. (2021). Stakeholder theory and management: Understanding longitudinal collaboration networks. *Plos one*, *16*(10), e0255658.
- Fares, J. (2023). A multi-level typology for stakeholder influence: A systematic literature review using the structural approach. European Management Journal. https://doi.org/10.1016/j.emj.2023.08.004

### 2.4. Comparative Analysis of Industry-Specific Challenges in Corporate Rankings:

In 1989, Barker et al. examined the challenges faced by Digital Equipment Corporation, identifying strategic/business, technical, and human resource/organizational challenges. Strategic/business challenges involved cross-functional business requirements that needed to be addressed through the implementation of configurators to improve business processes, which required support from top management. Technical challenges included the limitations of immature commercial configuration software, the difficulties of aligning the system with frequent product

upgrades and new product launches, the scalability of the system, and the size and complexity of the configurators. Finally, human resource/organizational challenges related to the awareness of key stakeholders and the need for organizational changes to support the implementation of the configurators. These challenges highlight the need for a comprehensive approach to addressing the various technical, organizational, and strategic factors that impact the successful adoption of new technologies.

- Kristjansdottir, K., Shafiee, S., Hvam, L., Forza, C., & Mortensen, N. H. (2018). The main challenges for manufacturing companies in implementing and utilizing configurators. *Computers in Industry*, *100*, 196-211.
- Barker, V. E., O'Connor, D. E., Bachant, J., & Soloway, E. (1989). Expert systems for configuration at Digital: XCON and beyond. *Communications of the ACM*, *32*(3), 298-318.

### 2.5. Cross-Sector Comparisons: A Tool for Assessing Company Rankings

Benchmarking across different industries and comparing performance can be a valuable learning and improvement tool for companies. By looking beyond their own industry, companies can gain new insights and ideas that they may not have discovered through traditional within-industry benchmarking. This approach can also help companies avoid stagnation and stay ahead of their competitors. By broadening their horizons and exploring new practices, companies can enhance their operations, improve efficiency, and ultimately achieve greater success. In short, cross-sector benchmarking can offer a fresh perspective and open up new opportunities for growth and improvement.

- Cankar, S. S., & Petkovsek, V. (2013). Private and public sector innovation and the importance of cross-sector collaboration. *Journal of Applied Business Research (JABR)*, *29*(6), 1597-1606.
- Walker, N. L., Styles, D., Gallagher, J., & Williams, A. P. (2021). Aligning efficiency benchmarking with sustainable outcomes in the United Kingdom water sector. *Journal of Environmental Management*, *287*, 112317.
- Walker, N. L., Styles, D., Coughlan, P., & Williams, A. P. (2022). Cross-sector sustainability benchmarking of major utilities in the United Kingdom. *Utilities Policy*, *78*, 101417.

## 3. Variable analysis

In the dataset, there are various variables present like name of the organization, type of industry, Country of origin, year founded, CEO of the organization, Revenue, Profits, Assets, Market Value, total employees, ranking of other organization in the year 2022.But we need to do descriptive statistics on qualitative variables that are revenue, profits, assets, market value and total number of employees.

- **Ranking**: Each organization's ranking for the current year is shown in this column. It shows where they are in relation to the other 2000 companies.
- **Organization Name**: The name of the business or establishment. This is an individual entry's unique identifier.
- **Industry**: explains the industry that the organization mainly works in. It gives background information on the company's area of expertise.
- **Country**: shows the nation of origin of every company. It is necessary to comprehend how these leading businesses are distributed throughout the world.
- **Year Founded:** This is the year that the organization was founded. This historical data can demonstrate the company's history and depth of experience.
- **CEO:** Stands for the organization's Chief Executive Officer. Gaining knowledge about the CEO might help one understand management and leadership.
- **Revenue (Billions):** The entire amount of money the business made this year, expressed in billions of units of currency
- **Profits (Billions):** Indicates the amount of money the business made this year. It displays the organization's financial prosperity.
- **Assets (Billions):** The total amount of the company's assets. Intangible assets like patents and trademarks are examples of intangible assets, as well as tangible ones like property and equipment.

## 4. Steps We used for Data Cleaning

- Converted data into .xlsx file from .csv file.
- Searched for duplicate rows and columns in the data to ensure unique data.
- Searched for special characters that are not needed in the data.
- Used trim function to remove extra spaces from the data.
- Searched for zero in the data and found zero in the cell in Total Employee column.
- Used excel built-in spelling checker to correct the incorrect spellings.
- Identified the number of missing values in all the columns in the data set.
- Removed square brackets from the cells in the Total Employee column.
- Then did the descriptive statistics for each quantitative variable like adjusted revenue, adjusted profits, adjusted assets, adjusted market value and total employees of the companies.

## 5. Descriptive Statistic measurements in Forbes Companies Ranking

We will utilize Excel to obtain the descriptive statistics for all of the data set. In order to determine the adjusted revenue, adjusted profit, adjusted assets, adjusted market value, and total employees, we will use descriptive statistics.

### 5.1. Descriptive Statistics for Adjusted Revenue:

*Table 1: Descriptive Statistics for Adjusted Revenue*

| *Adjusted Revenue (Billions)* | |
|---|---|
| | |
| Mean | 25.18662831 |
| Standard Error | 0.915929314 |
| Median | 13.47 |
| Mode | 2 |
| Standard Deviation | 40.9513625 |
| Sample Variance | 1677.01409 |
| Kurtosis | 43.51159873 |
| Skewness | 5.458611667 |
| Range | 572.75 |
| Minimum | 1.3 |
| Maximum | 574.05 |
| Sum | 50348.07 |
| Count | 1999 |

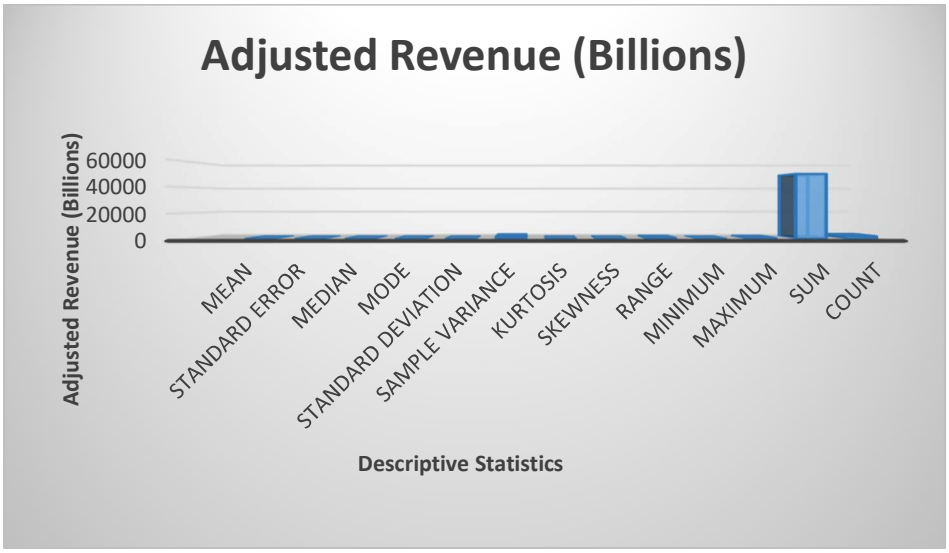

*Figure 1: A Graphical representation of Descriptive Statistics for Adjusted Revenue*

From the data table 1:

- The average of adjusted revenue for all the companies is $25.186 Billions.
- The Median for companies adjusted revenue is $13.47 Billion.
- The price values deviate from the average by $40.9513625 Billions. This indicates that the valuesare quiet variable. Some prices are extremely high while some are extremely low.
- The average of the squared differences between each data point and the mean (average) of the dataset is $98,683,552,495, this indicates that the dataset are quitespread out or varied. In other words, they don't cluster closely around the average; instead, they can be significantly above or below the average.
- The difference between the maximum revenue and minimum revenue is 572.75 Billions. This confirms high spreads and deviations in the data set.
- The maximum revenue is 574.05 and the minimum revenue is 1.3, this may indicate that some companies are facing issues in generating revenue as compared to other top revenue companies.

### 5.2. Descriptive Statistics for Adjusted Profit

*Table 2: Descriptive Statistics for Adjusted Profit*

| *Adjusted Profit (Billions)* | |
|---|---|
| | |
| Mean | 9.803166583 |
| Standard Error | 0.13778705 |
| Median | 8.36 |
| Mode | 8.2 |
| Standard Deviation | 6.160483486 |
| Sample Variance | 37.95155678 |
| Kurtosis | 115.5843961 |
| Skewness | 8.983825949 |
| Range | 117.41 |
| Minimum | -4.75 |
| Maximum | 112.66 |
| Sum | 19596.53 |
| Count | 1999 |

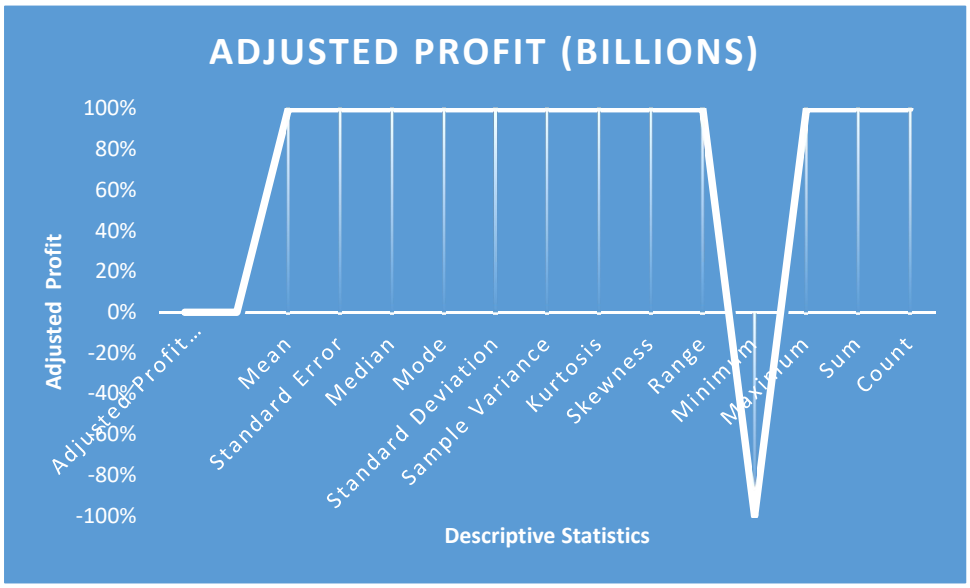


*Figure 2: A Graphical representation of Descriptive Statistics for Adjusted Profit*

From the above Data table 2:

- The average Mean of profit for all the companies is $9.803166583 Billions.
- The calculated mean for all the companies is $8.36 Billions.
- The profit deviate from the average by $6.160483486 Billions. This indicates that the values are quiet variable. Some companies have a profits that are extremely high while some have extremely low.
- The average of the squared differences between each data point and the mean (average) of the dataset is $37.95155678 Billions, this indicates that the dataset are quitespread out or varied. In other words, they don't cluster closely around the average; instead, they can be significantly above or below the average.
- The difference between the maximum revenue and minimum revenue is $572.75 Billions. This confirms high spreads and deviations in the data set.
- The maximum Profit is $112.66 and the minimum profit is $ -4.75, from table 2, we can evaluate that, the difference between the profits is relatively higher can usual .

This proves that some Companies don't make profits even being in the top list of the companies.

### 5.3. Descriptive Statistics for Adjusted Assets

*Table 3: Descriptive Statistics for Adjusted Assets*

| *Adjusted Assets (Billions)* | |
|---|---|
| | |
| Mean | 126.176013 |
| Standard Error | 8.137442815 |
| Median | 41.1 |
| Mode | 43.99 |
| Standard Deviation | 363.8265152 |
| Sample Variance | 132369.7332 |
| Kurtosis | 85.53517056 |
| Skewness | 8.282173369 |
| Range | 5518.34 |
| Minimum | 9.17 |
| Maximum | 5527.51 |
| Sum | 252225.85 |
| Count | 1999 |

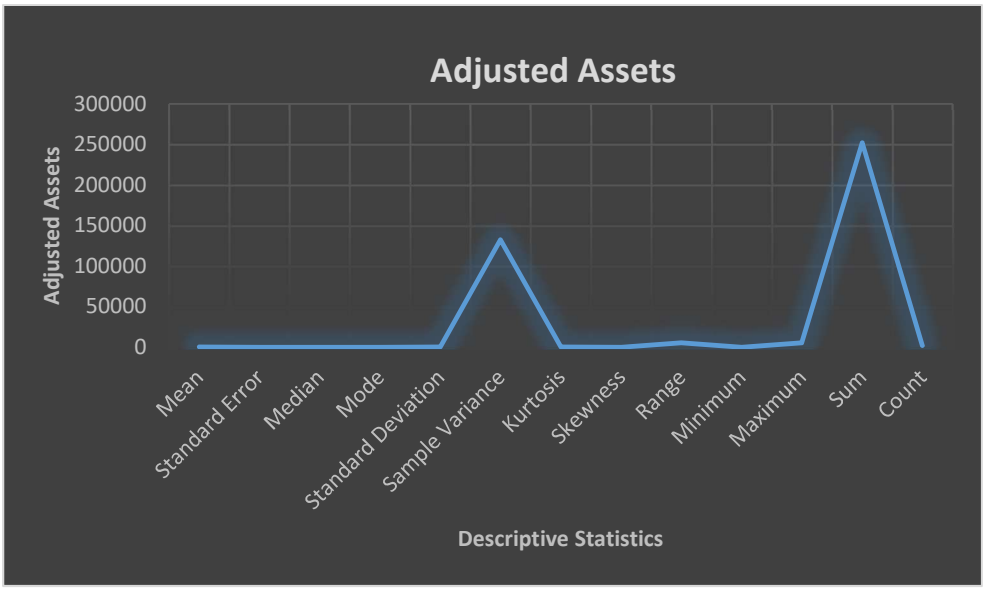


*Figure 3: A Graphical representation of Descriptive Statistics for Adjusted Assets*

From the above data table 3:

- The average Mean of assets owned by all the companies is $126.176013 Billions.
- The calculated median for all the companies in terms of assets is $41.1 Billions.
- The value of Standard deviation deviate from the average by $363.8265152 Billions. This indicates that the values are quiet variable. Some companies Owns large amount of assets.
- The average of the squared differences between each data point and the mean (average) of the dataset is $132369.73 Billions, this indicates that the dataset are quitespread out or varied. In other words, they don't cluster closely around the average; instead, they can be significantly above or below the average
- The difference between the maximum Assets owned and minimum Assets owned is 5518.34 Billions. This confirms high spreads and deviations in the data set.

### 5.4. Descriptive Statistics for Adjusted Market Value

*Table 4: Descriptive Statistics for Adjusted Market Value*

| *Adjusted Market Value (Billions)* | |
|---|---|
| | |
| Mean | 43.86067534 |
| Standard Error | 2.622071911 |
| Median | 22.24 |
| Mode | 8.06 |
| Standard Deviation | 117.2333014 |
| Sample Variance | 13743.64695 |
| Kurtosis | 261.9919114 |
| Skewness | 14.44449805 |
| Range | 2640.18 |
| Minimum | 5.74 |
| Maximum | 2645.92 |
| Sum | 87677.49 |
| Count | 1999 |

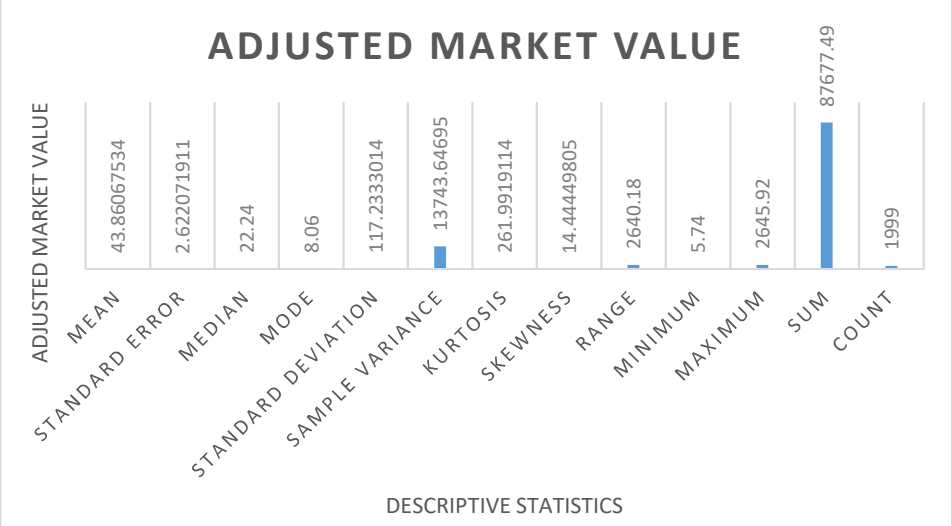


*Figure 4: A Graphical representation of Descriptive Statistics for Adjusted Market Value*

From the data table 4:

- The average Mean of Market Value for all the companies is $43.86067534 Billions.
- If we calculate Median of Market value for all the companies it comes to $22.24 Billions.
- The Standard deviation for Market value is $117.2333014 Billions. This indicates that the values are quiet variable. Some companies have a profits that are extremely high while some have extremely low.
- The average of the squared differences between each data point and the mean (average) of the dataset is $13743.64695 Billions, this indicates that the dataset are quitespread out or varied. In other words, they don't cluster closely around the average; instead, they can be significantly above or below the average.
- The difference between the maximum and minimum Market value is $2640.18 Billions. This confirms high spreads and deviations in the data set.
- The maximum market value is $2645.92 and the minimum market value is $5.74, from table 4, we can evaluate that, the difference between the Market value depends on the revenue and profit of your company.

### 5.5. Descriptive Statistics for Total employees

*Table 5: Descriptive Statistics for Total employees*

| *Total Employees* | |
|---|---|
| | |
| Mean | 48871.21861 |
| Standard Error | 2205.385665 |
| Median | 20507 |
| Mode | 0 |
| Standard Deviation | 98603.18521 |
| Sample Variance | 9722588134 |
| Kurtosis | 175.4056959 |
| Skewness | 9.901004886 |
| Range | 2300000 |
| Minimum | 0 |
| Maximum | 2300000 |
| Sum | 97693566 |
| Count | 1999 |

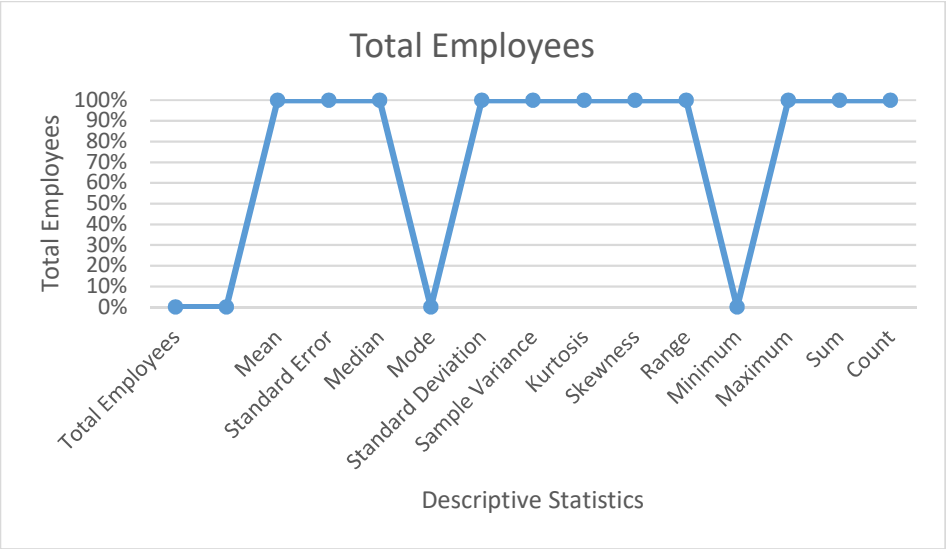


*Figure 5: A Graphical representation of Descriptive Statistics for Total Employees*

From the above Data table 5

- For the given dataset of the Total employees over the 1999 companies, the sum of all the total employees across all the companies, we can evaluate as Total figure of 97693566 employees.

- Considering the total number of employees across all the companies, the Average mean calculated of total employees in a company can be around 48871.218 employees.
- The Maximum number of employees registered in a single company that are employed are 2300000 employees.
- As there many big and small companies in the dataset provided, the deviated calculated is 98603.18521.

## 6. Result

In conclusion, Business analytics is significant for analyzing data, the statistics of companies of the world are useful for people who want to invest in these companies because the statistics provides valuable insights on adjusted revenue, adjusted profits, adjusted assets, adjusted market value and total employees of the companies. Trustworthy and thorough data of the companies is important for making financial decisions. Due to business analytics, large amount of data is converted in valuable insights to increase efficiency. People can use this data effectively and attain better results. Statistical data increases decision making quality. People with access to this data can assess risk, predict their investment performance and assess possible return of investment. After applying statistics on the cleaned data, we got concentrated data which included calculated evaluation of mean, standard error, standard deviation, median, variance, range, mode, sum, count etc. Then we did visualization and created visual representation of the calculative data and created line graph for profits and assets and bar graphs for the revenue and market value of the organizations.

Descriptive statistics and visualization provided base for deep analysis and decision making. As an outcome we received valuable information from the dataset like Standard deviation, Variance, range, mean, mode, median, sum etc. and converted it into different types of graphs that will help non-professional people to understand the data and make correct financial decisions.

## 7. Discussion:

Descriptive statistics are vital in research, enabling researchers to simplify and interpret complex datasets across disciplines. In epidemiology, they help analyse disease trends and identify risk factors, informing public health interventions. In education research, they assess student performance, teacher effectiveness, and educational policies. Clear data presentation, using measures like mean, median, and standard deviation, helps identify areas for improvement, guiding interventions to enhance education quality.

In business, descriptive statistics are essential for financial reporting, market analysis, and strategic decision-making. They provide decision-makers with a streamlined perspective of complex business data, empowering them to evaluate financial performance, market trends, and make well-informed choices. Descriptive statistics support studies in the social sciences by giving researchers the means to condense and analyse social processes.

Peer-reviewed publications explore the incorporation of Environmental, Social, and Governance (ESG) issues in firm evaluations within the context of corporate rankings. These studies show how important it is to take sustainability and ethical behaviour into account when evaluating companies, and they also show how important ESG aspects are for long-term financial performance and overall company rankings. Researchers investigate the complex relationship that exists between a company's reputation in the business community and its adherence to ESG principles. This relationship offers investors, policymakers, and corporate decision-makers insightful information.

The relationship between technology adoption and company outcomes is highlighted by recent study. Companies navigating the quickly changing technological world can benefit greatly from an understanding of technology acceptance and its implementation, which is essential for forecasting and assessing its impact on corporate success. This research gives organizations the information they need to make wise decisions and get the most out of adopting technology, which eventually helps them expand and compete in the market.

Research has shown that corporate rankings are heavily influenced by stakeholder views. These perceptions and a company's standing are intricately linked. Scholars examine the perspectives of different stakeholders, such as clients, staff, and the local community, on a business. By emphasizing the crucial role that stakeholder connections and reputation play in fostering business success, this approach contributes to our knowledge of the non-financial factors that affect corporate rankings.

Thorough analyses of industry-specific issues in company rankings offer crucial insights into the distinct problems encountered by different industries. Researchers study factors causing ranking variations across industries, deepening understanding of unique business challenges. By making it easier to develop customized plans and interventions to address issues unique to a given industry, this comparative method encourages the development of more precise and nuanced

assessments of company performance. This research allows for a more focused and efficient strategy to raising business performance and rankings by identifying the unique obstacles faced by each industry.

The effectiveness of cross-sector comparisons in evaluating company rankings has been demonstrated by recent studies. Researchers offer a comprehensive understanding of company performance by the adoption of a cross-sector approach, which facilitates a profound comprehension of elements that surpass industry-specific constraints and impact overall rankings. Comparative analyses of this kind aid in the creation of flexible frameworks that may be used to assess corporate success across a range of industries.

## References:


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   https://www.forbes.com/sites/forbesstaff/2022/05/12/forbes-global-2000-list-2022-the-top-200/?sh=6e834ede3290

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## Self Reflection

As I was assigned underway on this fantastic task, I realized that I needed to learn more about the statistics. I explored resources for a reliable data source for our Literature Review. Using my research skills, our team collaborated on various articles and journals. My task is to examine descriptive statistics within the Forbes dataset provided. This approach will help us understand the data and its impact on our research.

My analytical work has enhanced my ability to analyse data effectively. Working on this innovative forecasting project has been exhilarating. As a novice, I was captivated by the unexpected challenges that emerged from the data, providing valuable learning experiences. This hands-on approach has broadened my understanding of the complexities involved in data analysis.

In my work, I analysed the Forbes 2022 company rankings data, which included metrics such as adjusted revenue, profit, assets, market value, and total employees. I thoroughly cleaned the data and used Excel's data analysis tab to calculate descriptive statistics for each variable in the dataset. This rigorous process ensured the accuracy and integrity of the data, allowing for a comprehensive analysis of the company rankings.

My analysis included data cleaning, statistics, and visualization. I calculated mean, median, mode, and standard deviation. Our team created clear charts and graphs for effective visualization. This cooperation helped us present company rankings comprehensibly.

In conclusion, business analytics extracts valuable insights from data, enhancing efficiency. Descriptive statistics and visualization are essential for informed decision-making and business success. Organizations can use these tools to make data-driven decisions that improve operations and achieve objectives.

I aim to improve my business statistics skills to exceed standard analysis quality. This will help me derive valuable insights from data and contribute to data-driven decision-making in the business world. Continuous learning will enable me to stay ahead and deliver exceptional results.